

EEG-Based Analysis of Alpha Brain Activity During Eyes-Open and Eyes-Closed Conditions

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Abstract

Electroencephalography (EEG) is a technique used to record the electrical activity of the brain. This project investigates the differences in alpha-band power between eyes-open and eyes-closed resting conditions using an eight-channel EEG dataset.

The analysis was performed using Python. EEG signals were collected from eight channels (EXG 1–EXG 8) at a sampling frequency of 250 Hz. The signals were pre-processed using a Butterworth bandpass filter with a frequency range of 0.5–40 Hz and a 50 Hz notch filter. Power spectral density was estimated using Welch's method. Alpha-band power was calculated within the 8–12 Hz frequency range using non-overlapping five-second windows.

The results showed that mean alpha power was higher during the eyes-closed condition than during the eyes-open condition across all eight EEG channels. The highest mean alpha power during eyes closed was observed in EXG 1, with a value of $88.276 \mu V^2$, while the lowest was observed in EXG 6, with a value of $34.235 \mu V^2$.

These findings demonstrate a consistent difference in alpha-band power between the two recording conditions. The observed pattern is consistent with the commonly reported increase in alpha activity during eyes-closed resting conditions. However, the analysis is based on one recording per condition, and the findings should therefore be interpreted as a descriptive comparison rather than evidence of population-level statistical significance.

1.Introduction

The human brain continuously produces electrical activity through the communication of neurons. This activity can be recorded using a technique known as Electroencephalography (EEG).

EEG is a non-invasive method that measures electrical signals generated by the brain through electrodes placed on the scalp. It is widely used in neuroscience and brain-computer interface (BCI) research to study brain activity under different conditions.

EEG signals contain different frequency components, commonly classified into frequency bands such as delta, theta, alpha, beta and gamma. Each frequency band represents a different range of oscillatory activity.

Alpha waves generally occur within the frequency range of 8–12 Hz. Alpha activity is commonly observed during relaxed wakefulness and is often more prominent when a person closes their eyes compared with when their eyes are open.

In this project, EEG recordings from eyes-open and eyes-closed resting conditions are analysed to investigate differences in alpha-band power across eight EEG channels. Python is used for signal preprocessing, frequency-domain analysis, alpha-power calculation and visualisation.

The project aims to understand how alpha activity changes between the two resting conditions and to demonstrate the use of digital signal-processing techniques in EEG research.

2.Objectives

The main objectives of this project are:

1. To understand the fundamentals of Electroencephalography (EEG) and its applications in brain-computer interface research.
2. To analyse EEG signals recorded under eyes-open and eyes-closed resting conditions.
3. To preprocess EEG signals using digital filtering techniques.
4. To calculate the Power Spectral Density (PSD) using Welch's method.
5. To extract alpha-band power within the frequency range of 8–12 Hz.
6. To compare alpha power across eight EEG channels under both recording conditions.
7. To visualise and interpret the differences in alpha activity between eyes-open and eyes-closed conditions.
8. To understand the application of Python in EEG signal processing and analysis.

3. Dataset Description

The EEG dataset used in this project was obtained from the Unicorn BCI Core 8 EEG Datasets repository on GitHub.

The dataset contains EEG recordings collected under different experimental conditions. For this project, two resting-state conditions were selected:

- **Eyes-Open Condition:** EEG signals recorded while the subject's eyes were open.
- **Eyes-Closed Condition:** EEG signals recorded while the subject's eyes were closed.

3.1 EEG Recording Specifications

The following specifications describe the EEG recordings used in this project:

Parameter	Description
EEG Device	Unicorn BCI Core 8
Number of EEG Channels	8
Sampling Frequency	250 Hz
Eyes-Open Recording	1 recording
Eyes-Closed Recording	1 recording
Total EEG Channels Analysed	8
Alpha Frequency Range	8–12 Hz

3.2 EEG Channels

The eight EEG channels used in the analysis are:

EXG 1 – Fz

EXG 2 – C3

EXG 3 – Cz

EXG 4 – C4

EXG 5 – Pz

EXG 6 – P07

EXG 7 – P0z

EXG 8 – P08

3.3 Data Format

The EEG recordings were provided in CSV (Comma-Separated Values) format. The files contain eight EEG signal columns along with additional recording information.

The EEG data were loaded into Python using the Pandas library for further preprocessing and analysis.

4. Methodology

The methodology describes the steps followed to process and analyse the EEG recordings under eyes-open and eyes-closed conditions.

4.1 Data Loading

The EEG recordings were loaded from CSV files using the Pandas library in Python.

The eyes-open and eyes-closed recordings were loaded separately.

The eight EEG channels (EXG 1 to EXG 8) were selected for further analysis.

4.2 Data Quality Assessment

The dataset was examined to check for missing values, infinite values and recording-quality indicators. The VALID and SATURATION columns were also inspected to assess the quality of the recorded signals.

4.3 Signal Preprocessing

The EEG signals were filtered to reduce unwanted frequency components and noise.

The following filtering techniques were used:

- **Butterworth Bandpass Filter:** A fourth-order Butterworth filter with a frequency range of 0.5–40 Hz was applied.
- **Notch Filter:** A 50 Hz notch filter was additionally applied as part of the preprocessing pipeline. Because the preceding bandpass filter was limited to 0.5–40 Hz, the 50 Hz component was already strongly attenuated by the bandpass stage.

Zero-phase filtering was performed using forward-backward filtering functions.

4.4 Power Spectral Density Analysis

Power Spectral Density (PSD) was calculated using Welch's method to examine the distribution of signal power across different frequencies.

The following parameters were used:

- Sampling frequency: 250 Hz
- Window duration: 5 seconds
- Segment length: 1250 samples
- Window overlap: 1125 samples
- Step size: 0.5 seconds

4.5 Alpha-Band Power Extraction

Alpha-band power was calculated within the frequency range of 8–12 Hz.

The filtered EEG signals were divided into non-overlapping five-second windows. Welch's method was applied to each window, and the power within the alpha frequency range was integrated to obtain alpha-band power.

4.6 Statistical and Descriptive Analysis

For each EEG channel, the following descriptive measurements were calculated:

- Mean alpha power
- Standard deviation of alpha power
- Number of analysed windows

The results from the eyes-open and eyes-closed conditions were compared across all eight channels.

4.7 Data Visualisation and Result Storage

Graphs were generated using Matplotlib to visualise the EEG signals, power spectral density and alpha-power comparisons.

The calculated results were saved as CSV files for future reference and inclusion in the final project report.

5. Results and Analysis

After preprocessing the EEG signals and calculating alpha-band power, the results from the eyes-open and eyes-closed conditions were compared across all eight EEG channels.

5.1 Alpha Power Comparison

The mean alpha power was calculated for each channel under both recording conditions.

Channel Mean Alpha Power – Eyes Open (μV^2) Mean Alpha Power – Eyes Closed (μV^2)

EXG 1	7.4275	88.2761
EXG 2	7.5614	62.5520
EXG 3	7.4387	76.5330
EXG 4	4.1368	40.7754
EXG 5	6.9118	51.6177
EXG 6	4.5859	34.2347
EXG 7	5.9465	43.5447
EXG 8	2.7970	42.3830

5.2 Statistical Summary

The mean and standard deviation of alpha power were calculated using the analysed five-second windows.

There were 60 windows in the eyes-open condition and 61 windows in the eyes-closed condition for each channel.

The standard deviation was higher during eyes closed across all eight channels, indicating greater variation in alpha power across those windows.

Channel SD Eyes Open (μV^2) SD Eyes Closed (μV^2)

EXG 1	2.9898	35.7978
EXG 2	3.1692	23.5732
EXG 3	2.8006	31.2785
EXG 4	1.5938	17.2140
EXG 5	2.8274	22.4915
EXG 6	2.2414	15.4559
EXG 7	3.3905	17.9520
EXG 8	0.9938	13.3099

5.3 Alpha Power Difference

The difference between the mean eyes-closed and eyes-open alpha power was calculated for each channel.

Channel Alpha Power Difference (μV^2)

EXG 1	80.8487
EXG 2	54.9906
EXG 3	69.0943
EXG 4	36.6386
EXG 5	44.7058
EXG 6	29.6487
EXG 7	37.5983

Channel Alpha Power Difference (μV^2)

EXG 8 39.5860

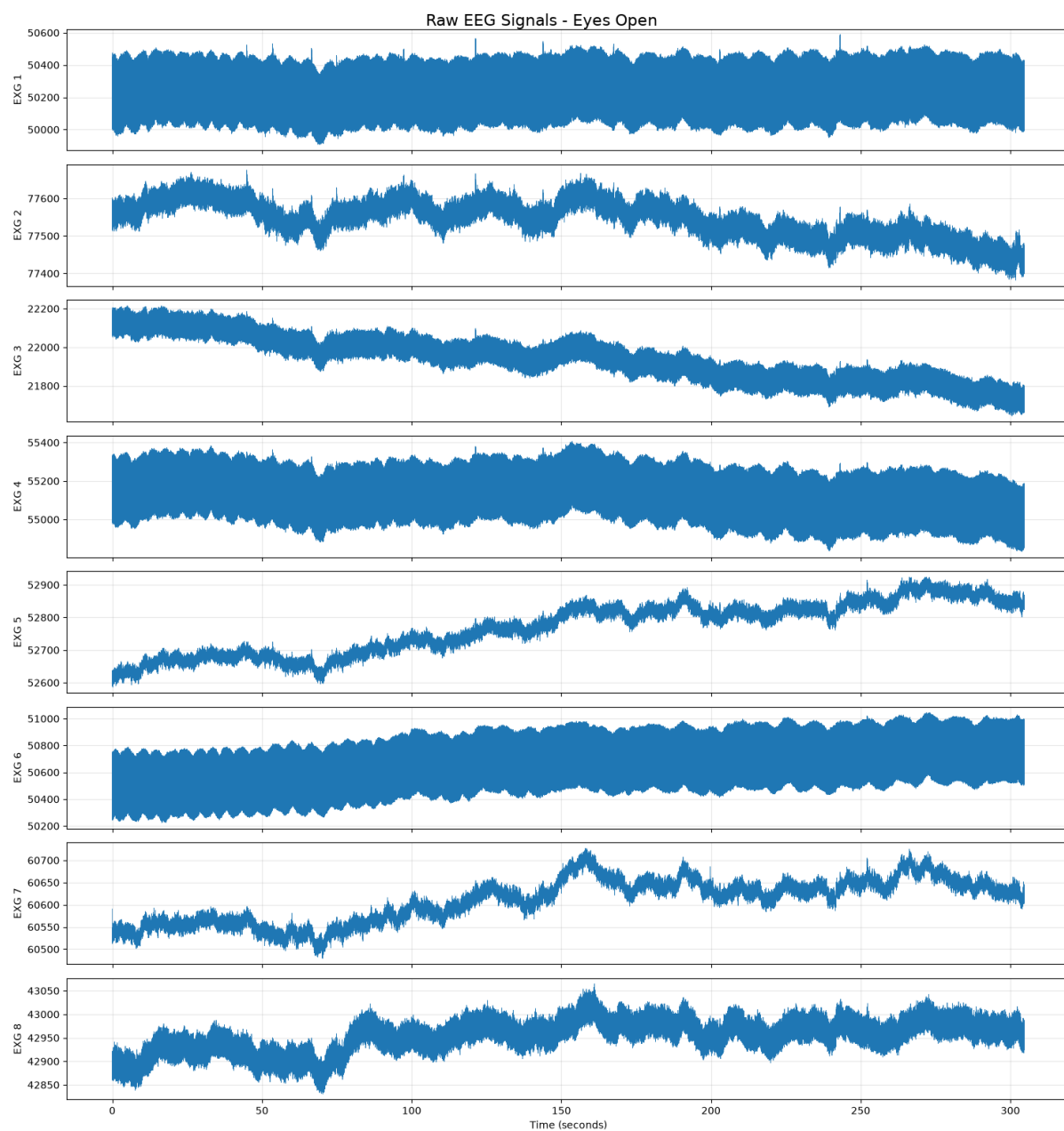
5.4 Interpretation of Results

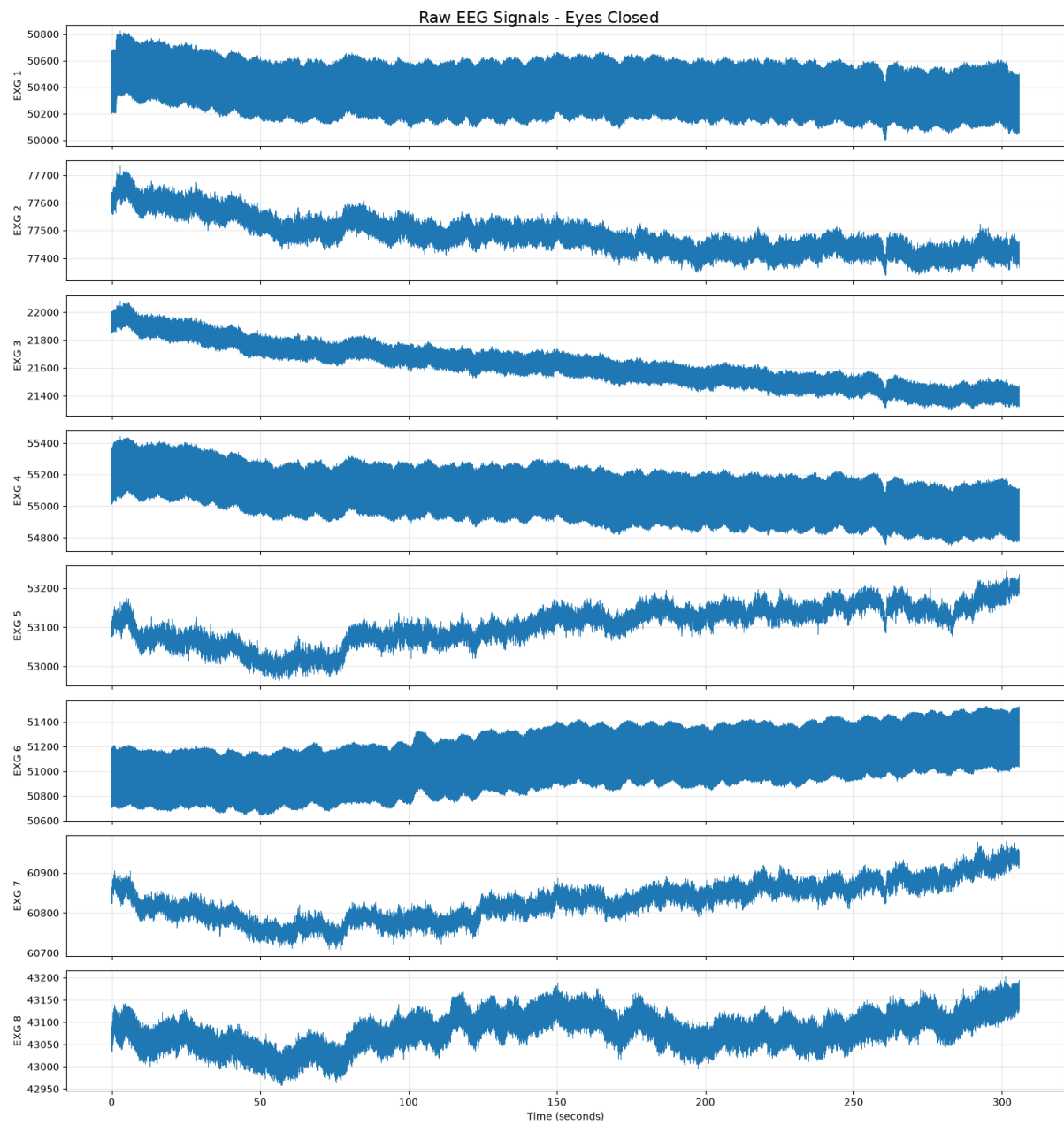
The results show that mean alpha power was higher during the eyes-closed condition than during the eyes-open condition across all eight EEG channels.

EXG 1 showed the largest difference in mean alpha power, while EXG 6 showed the smallest difference.

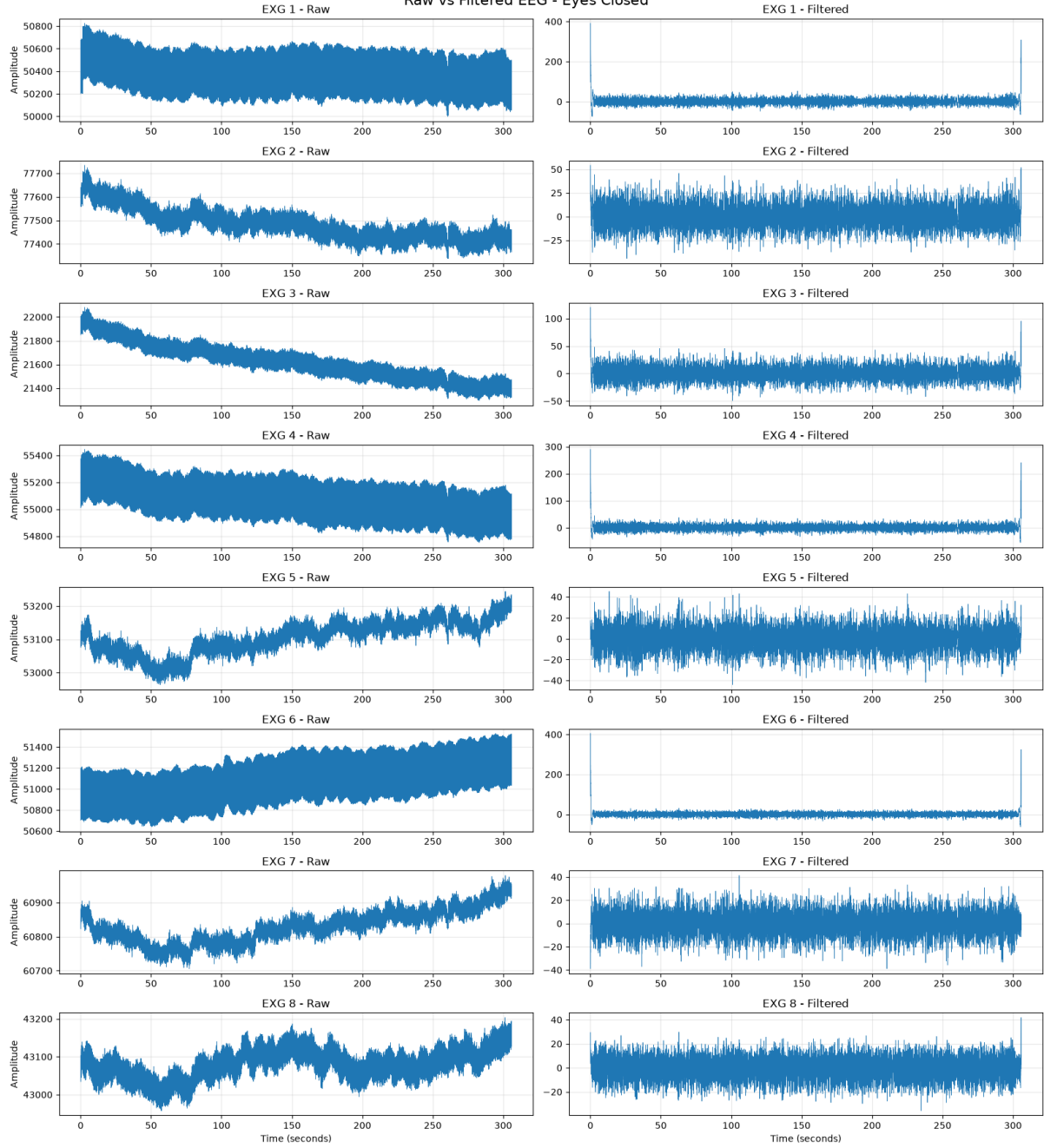
This observation is consistent with the commonly reported increase in alpha activity during eyes-closed resting conditions.

The results represent a descriptive comparison of the two recordings and do not establish statistical significance across a wider population.

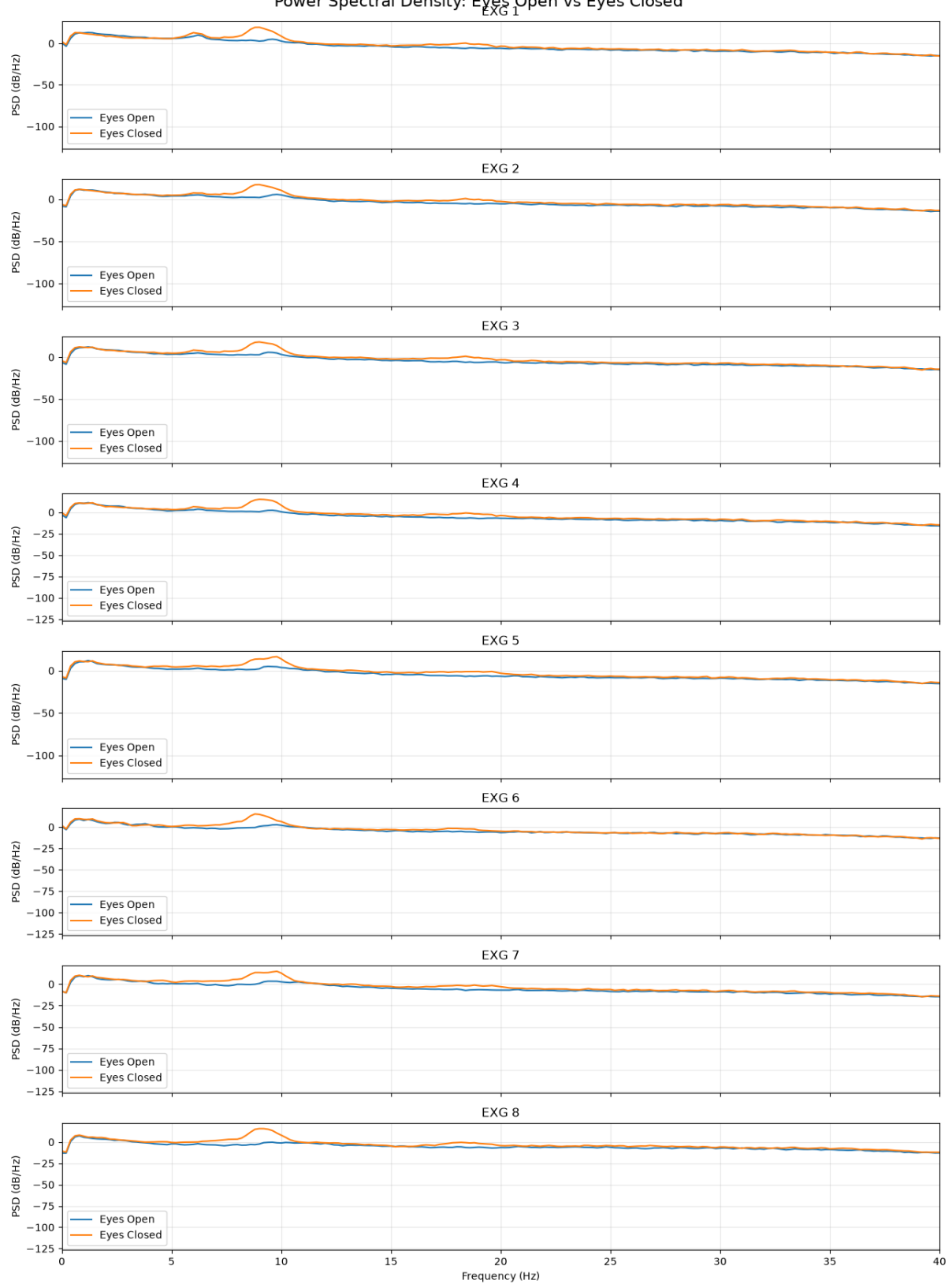


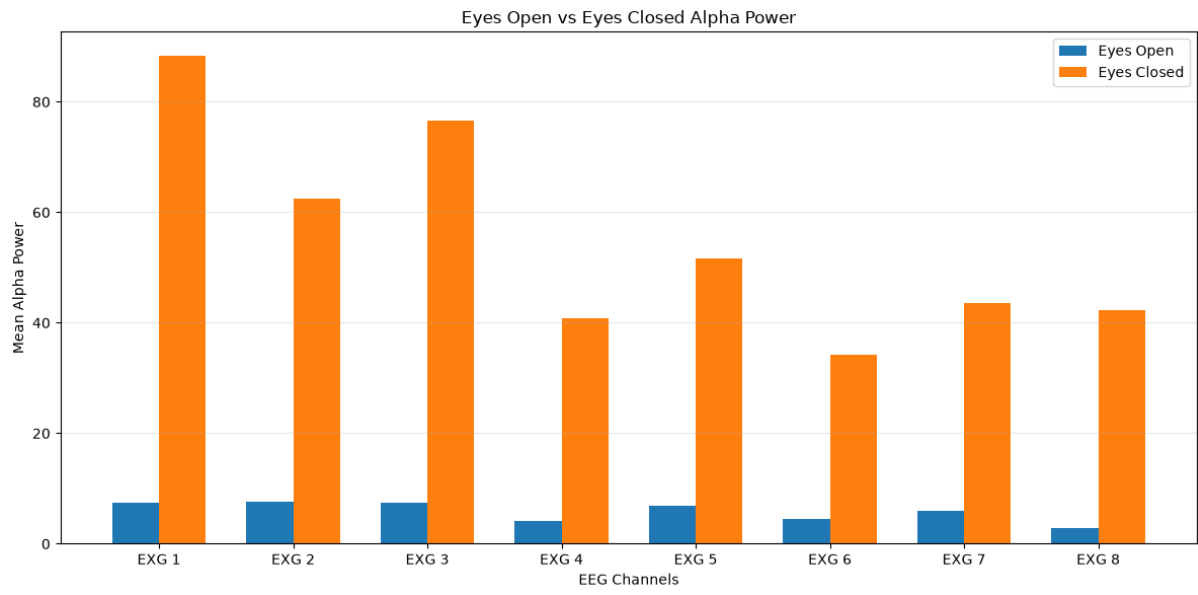


Raw vs Filtered EEG - Eyes Closed

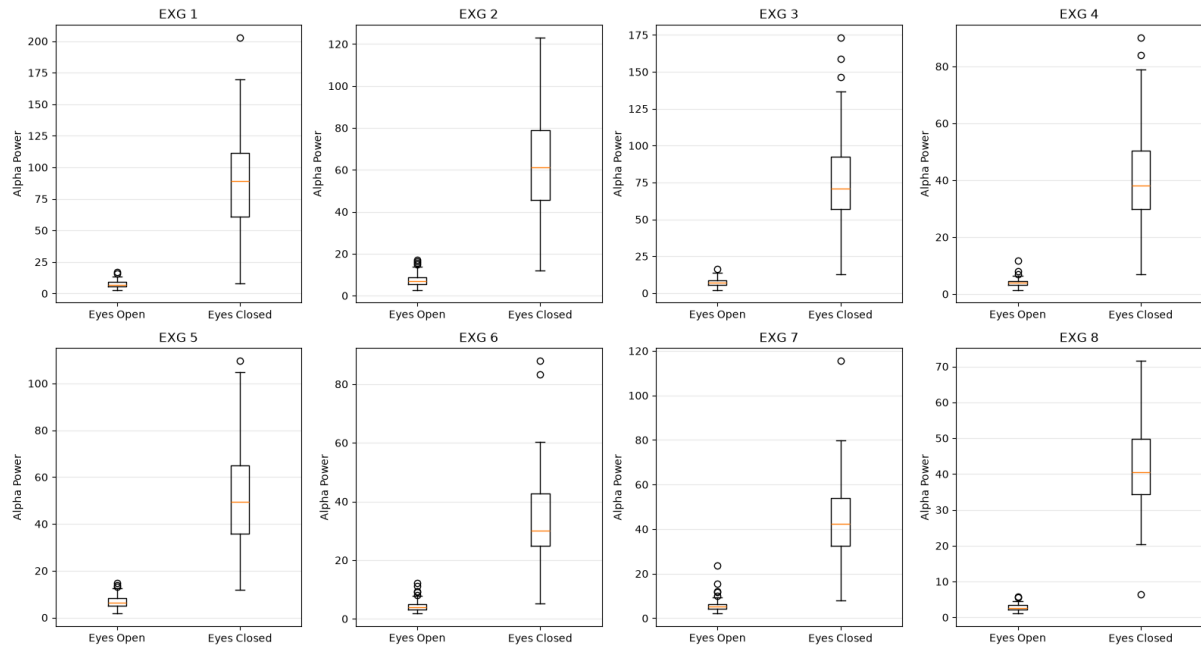


Power Spectral Density: Eyes Open vs Eyes Closed





Alpha Power Distribution: Eyes Open vs Eyes Closed



6. Discussion

The purpose of this study was to investigate the differences in alpha brain activity between eyes-open and eyes-closed conditions using an eight-channel EEG dataset.

The results showed that mean alpha power was higher during the eyes-closed condition across all eight EEG channels. This observation is consistent with the commonly observed increase in alpha activity when the eyes are closed and visual input is reduced.

The Power Spectral Density (PSD) analysis helped examine the frequency distribution of the EEG signals, while the alpha power analysis provided a quantitative comparison within the 8–12 Hz frequency band.

The boxplots further illustrated the differences in alpha power distributions between the two recording conditions. The standard deviation was also higher during the eyes-closed condition across all eight channels, indicating greater variation in alpha power across the analysed windows.

However, the findings should be interpreted carefully because the analysis used only one recording for each condition. The results describe the observed recordings and cannot establish that the same pattern will occur in every individual.

Overall, this study demonstrates how EEG preprocessing, frequency-domain analysis, and alpha-band power estimation can be used to examine changes in brain activity between eyes-open and eyes-closed conditions.

7. Limitations of the Study

Although this study provided useful observations about alpha brain activity under eyes-open and eyes-closed conditions, it has some limitations.

1. Limited Number of Recordings

The study used only one EEG recording for each condition. Therefore, the results cannot represent the responses of a larger group of individuals.

2. Limited Number of EEG Channels

The analysis was performed using eight EEG channels. Additional channels could provide more detailed information about brain activity across different scalp regions.

3. Possible Noise and Artifacts

EEG signals can contain unwanted noise caused by eye movements, muscle activity, and environmental interference. Although preprocessing filters were applied, these filters alone cannot remove every artifact.

4. Limited Recording Conditions

The analysis focused only on eyes-open and eyes-closed conditions. Other conditions, such as different mental tasks or levels of attention, were not investigated.

5. Limited Generalization

The findings describe the two recordings analysed in this project. Further recordings involving multiple participants and repeated trials would be required to determine whether the observed pattern is consistent across individuals.

Despite these limitations, the study provides a practical demonstration of EEG signal preprocessing and alpha-band power analysis.

8. Conclusion

This research project investigated alpha brain activity under eyes-open and eyes-closed conditions using an eight-channel EEG dataset.

The EEG signals were analysed using Python. The preprocessing stage included bandpass filtering between 0.5 and 40 Hz and notch filtering at 50 Hz to reduce unwanted frequency components. Power Spectral Density (PSD) analysis was then performed, followed by the calculation of alpha-band power within the 8–12 Hz frequency range.

The results showed that mean alpha power was higher during the eyes-closed condition than during the eyes-open condition across all eight EEG channels. This observation is consistent with the commonly observed increase in alpha activity during eyes-closed resting conditions.

The project demonstrates how EEG signal processing techniques, including filtering, PSD estimation, and frequency-band power analysis, can be used to study differences in brain activity.

Although the study was limited to one recording per condition, it provides a foundation for further research involving more participants, repeated recordings, and additional EEG analysis techniques.

In conclusion, the project successfully demonstrated the use of Python-based EEG analysis to compare alpha brain activity between eyes-open and eyes-closed conditions.

9. References

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